POKHARA UNIVERSITY

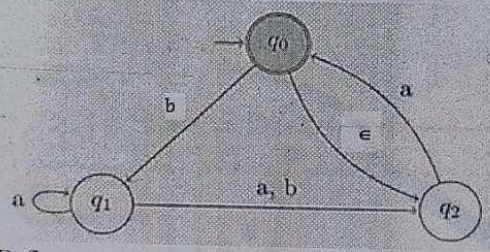
Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks:100
Course: Theory of Computation | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is function? Explain its types with examples. 7
   b) Differentiate between DFA and NFA? Design a DFA that accepts the language given by L={w∈{a,b}* : w contains neither 'aa' nor 'bb' as substring}. Hence test your design for abaabb. 8
2. a) Convert the following NFA to its equivalent DFA. 7



   b) Define pumping Lemma for regular language. Show that $L=\{a^n b^{2n}: n>1\}$ is not regular using pumping lemma for regular language. 8
3. a) What is CFG? Design CFG for language $L=\{a^m b^n : m>=1, n>=1\}$. Test the grammar for derivation of aaaabbb and also draw equivalent parse tree. 8
   b) Convert the following grammar into Chomsky Normal form. 7

   $S \rightarrow bAD$
   $A \rightarrow aB/bAB$
   $B \rightarrow b$
   $D \rightarrow \varepsilon$ (Null)
4. a) Define PDA with block diagram. Design a PDA which accepts the language $L=\{a^{4n} : n>=1\}$ and test for strings aaaaaaaa and aaaaaa. 8

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b) Show that the language L= $\{a^n b^n c^n : n>0\}$ is not context free using the concept of pumping lemma. 7

5. a) Define Turing machine. Design a Turing machine that accepts the language L= $\{a^n b^n c^n : n>=0\}$. 8

b) How does a Turing machine compute a function of natural numbers? Describe. Show that the function $f(n) = n + 2$ is computable. 7

6. a) State the halting theorem and give the outline of its proof. 7

b) What are P, NP and NP-Complete problems? Explain with examples. 8

7. Write short notes on: (**Any two**) 2×5

a) Simplification of CFG

b) Recursive and Recursively Enumerable Language

c) Decision algorithm for CFL

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